CHAPTER - 3

SEMI - CONDUCTING MATERIALS.

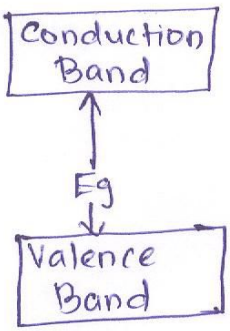
Energy band diagram.

Energy band diagram consist of two bands

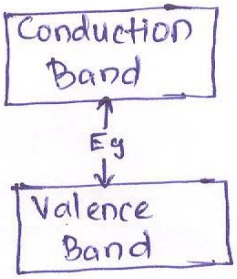
-> Conduction band

-> Valence band.

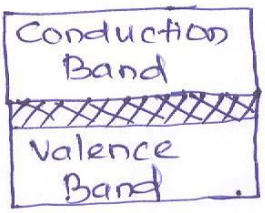
These two bands separated by forbidden energy gap Eg.



(a) Insulator



(b) Semi-Conductor



(c) Conductor.

The energy band diagram of Insulator, Semi-conductor and Conductor are shown in above figure. The lower band filled with electrons and upper band is empty.

The forbidden energy gap of insulator is 5 to 10 eV. and energy gap of semi conductor is 0.20 to 2.5 eV. There is no energy gap between conduction band to valence band for conductor. i.e two bands are overlap with each other.

An insulator and Pure semi conductor the conduction takes place only when the valence band electrons are jumps to conduction band. Due to large energy gap in insulator the electrons in valence band cannot jump to conduction band.

Therefore conduction is not possible in the insulator.

In Semi-conductor, with increase temperature may electrons move from valence band to conduction band.

In conductors the two bands were overlap with each other, so little bit temperature is enough to move electrons from valence to conduction band.

Semi-conductors.

Semi conductors are materials which have electrical conductivity lying between good conductor and insulator. The resistivity of semi-conductor varies from 10^{-8} to 10^{-6} ohm. (for conductor - 10^{7} to 10^{8} ohm). The general semi conducting materials are

→ Germanium
→ silicon

Another important characteristics of semi-conductor is that they have small energy gap (0.2 to 2.5eV) The Semi-conductors are classified as follows.

→ Intrinsic semi conductors.
→ Extrinsic semi conductors

(i) Intrinsic Semi-conductors.

A pure semi-conductor is called intrinsic semi-conductor. Here no free electrons are available. The valence band filled with electrons and conduction band is empty, these bands are separated by energy gap Eg.

[K - Kelvin Unit for temperature]

At '0' k, the electric conduction is not possible. A Pure Semi-conductor behaves as an insulator. The resistance of a Semi conductor decreases when increase in temperature. when an electric field applied to Intrinsic Semi conductor at a temperature greater than 0°K, the electrons move from valence band to conduction band hence the hole are created in the valence band.

In Intrinsic Semi conductor electrons in the Conduction band is equal to holes' in the valence band.

$$n = p$$

(ii) Exterinsic Semi Conductor

An amount of impurities added with pure Semi conductor is called exterinsic semiconductor. The Process of adding impurities is called doping. Generally the following doping agents are used

→ Pentavalent atoms having 5 valence electrons called donar atoms. (arsenic, antimony, Phosphorus)

→ Trivalent atoms having 3 valence electrons called acceptor atoms. (gallium, aluminium, boron)

Donar impurities donates their valence electrons to a Semi-conductors conduction band Providing excess electrons to the Intrinsic semi conductor. Excess electrons Increases the carrier concentration of the semiConductor making it n-type.

Acceptor impurities have fewer valence electrons. They accept electrons from valence band. This Provides excess holes to the Intrinsic semi Conductor. Excess holes increases the hole carrier concentration to make P-type.

Two types of extrinsic Semi Conductor

(i) N-type Semi Conductors.
(ii) P-type Semi Conductors.

(i) N-type Semi Conductors.

Extrinsic semi Conductor with a large electron Concentration than hole concentration is known as N-type semi Conductor. The Pharse 'n-type' comes from the negative charge of the electron. In n-type semi-Conductor electrons are the majority carriers and holes are minority carriers.

(ii) P-type Semi-Conductors.

P-type semi conductor have large hole Concentration The Phrase 'P-type' comes from positive charge of the hole. In P-type semi conductor holes are majority carriers and electrons are minority carriers.

Drift and Diffusion current.

Drift Current

A Semi conductor may be doped with impurity atoms so that the current is due to either electron or holes to the metal where current carriers are electrons only.

The transportation of charge under the influence of an electric field is called Drift Current.

~~Diffusion current~~. Let us consider a conductor of length L containing N electrons. If a constant electric field E is applied to the conductor the electron would be accelerated with acceleration.

$a = \frac{qE}{m}$ $q \rightarrow$ charge of carrier

$V = \mu E$ $V \rightarrow$ average speed of carrier

$\mu \rightarrow$ mobility of charge carrier

Now a charge carrier q takes time t to travel a distance L in the conductor, the total number of electrons passing through cross section A of the wire in unit time will be $\frac{N}{t}$.

So the total charge per second passing any area is the current I is given by

$$I = \frac{Nq}{t} = \frac{NqV}{L}$$ because $\frac{L}{t}$ is the average velocity

If J is the current density, then

$$J = \frac{I}{A} = \frac{NqV}{LA} \Rightarrow \frac{N}{LA}$$ represents charge carrier

$$\boxed{J = nqV}$$

Diffusion current.

The transportation of charge under the influence of an electric field as a results of non-uniform concentration gradient is called diffusion current.

This diffusion current is not encountered in metals

Let us consider non-uniform concentration of ~~carriers~~ holes 'p' varies with distance 'x' in the semiconductor and exist of concentration gradient $\frac{dP}{dx}$.

Current density

$J_p = -q D_p \frac{dP}{dx}$ (holes)

$D_p \rightarrow$ Diffusion constant

$J_n = q D_n \frac{dn}{dx}$ (electron

The total current will be sum of Drift and diffusion current

holes.

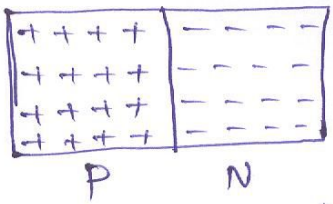
$$J_p = Pq\mu_p E - q D_P \frac{dP}{dx}$$

electrons.

$$J_n = nq\mu_n E + q D_n \frac{dn}{dx}$$

P-N Junction.

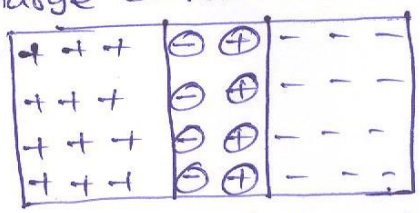
It is possible to manufacture a single piece of semi conductor material half of which is doped by P-type impurity and the other half by N-type impurity.



This P-N Junction is fundamental to the operations of diode transistors and other solid state devices.

P-region consists of excess holes and N-region consists of excess electrons. The Property of atom causes to flow from high concentration region to low concentration region. This force causes the electrons to diffuse into P-region and the holes to diffuse into N-region.

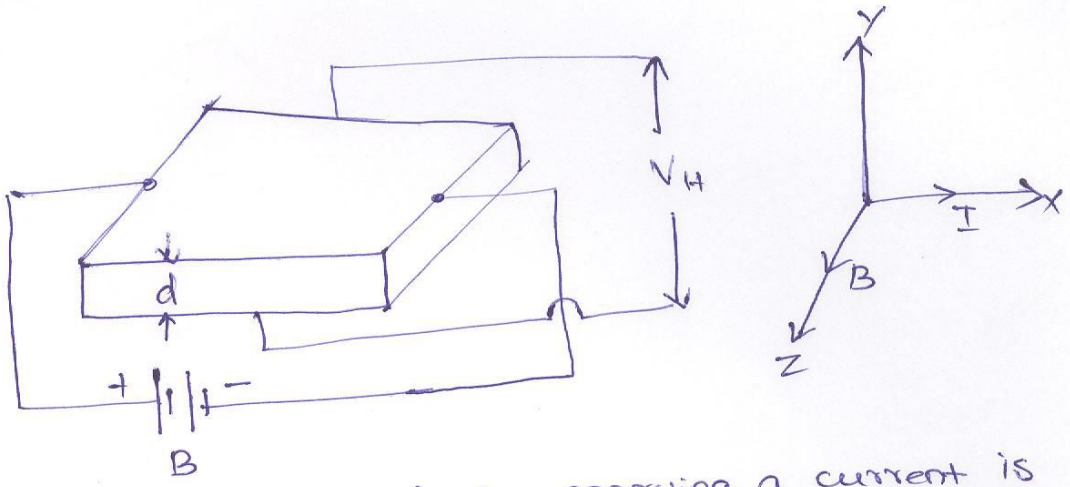
In the P-region the excess charge carries are holes and when an electron cross the junction it falls into a hole and stays in the layer. Similarly when a hole diffuse into n-region it recombines with a hole and stay in the layer.



Due to this causes a region near to Junction which does not contain any free charge carriers and therefore is known as depletion region.

When the electron diffuse into P-region it leaves a positively charged ion in the n-region and after recombination a negatively charged ion is created in the P-region.

Hall effect.



If a piece of conductor carrying a current is placed in a magnetic field, an electric field is produced inside the conductor in a direction normal to both current and the magnetic field. This phenomenon is known as Hall effect and the generated electric field is known as Hall field.

Hall effect is used for the following purposes

→ To determine semi conductor type (P or N)
→ To find carrier concentration.
→ To measure the conductivity of the material
→ To find carrier mobility.

Consider a rectangular piece of conductor carrying a current I, in the positive x direction and subjected to the magnetic flux density in the positive z. direction. If the current carriers are electron, as in the case of an n-type semi conductor, this accumulation will make the bottom surface negatively charged with respect to top surface. Therefore an electric field is called Hall field, will be developed along the negative y-direction. At this stage no further accumulation of electrons takes place on the bottom surface and Hall field reaches a steady value.

If the current carrier are holes i.e when the Conductor type is P-type, the accumulation of carriers on the bottom surface will make this surface positively charged with respect to top surface. In this case the hall field is produced along the positive Y-direction. At this stage no further accumulation of holes takes place on the bottom surface and Hall field reaches a steady value.

If E_H is the Hall field in the Y-direction

$$eE_H = eVB \quad \text{—(1)}$$

$$E_H = VB$$

If J is the current density in x-direction

$$J = n_c eV \quad \text{—(2)}$$

n_c → Concentration of current carrier

From eqn (1) and (2)

$$E_H = \frac{JB}{n_c e}$$

If the Hall effect described by the hall Coefficient R_H defined in terms of current density J

$$R_H = \frac{1}{n_c e} \quad \text{so} \quad E_H = R_H J B$$

If the current carriers are electrons $R_H = \frac{1}{ne}$

If the current carriers are holes $R_H = \frac{1}{pe}$

where p & n → represents hole and electron Concentration.

The sign of charge carrier
- Electron – Negative
- Holes – Positive.

Thus the sign of the Hall Co-efficients tells us whether the sample is an n-type or P-type Semi conductors.

Difference between Semi Conductors to Conductors

Semi conductors	Conductors
(i) Their resistivity high and temperature coefficient of resistance is always negative	The resistivity is low and temperature coefficient of resistance is not constant.
(ii) These are material having small forbidden energy gap	These materials overlapping energy band.
(iii) At low temperature they become dielectric	At very low temperature. they become super conductors.
(iv) Both resistivity and contact effects are very sensitive to small change in Physical Conditions.	Both resisitivity and contact effects are very sensitive to small change in physical Conditions.
(v) They show a rise in conductivity with increase in temperature	The conductivity increases with decreasing temperature.

Difference between Semi conductors to Insulators.

Semi conductors	Insulators.
(i) These are the materials whose valence electrons are bonded somewhat loosely to their atoms.	These are the materials which valence electrons used are bonded very tightly to their atoms.
(ii) They require less energy than insulator and more than Conductors to move electrons.	Forbidden energy gap is too large.
(iii) They have (i) Empty in conduction band (ii) almost filled valence band (iii) Energy gap 0.2 to 2.5 ev	They have (i) Empty in Conduction band (ii) Almost filled valence band (iii) Energy gap 5 to 10 eV.

Comparision of Germanium and Silicon.

Germanium	Silicon
(1) Transistor action was first observed. Pure form of Germanium available	Next to Germanium. Its need to Purify to get Pure form of silicon.
(2) Low melting Point	High melting Point (1420°C)
(3) This material has low sensitivity	This material had high sensitivity due to presence of oxygen (10^{17} to 10^{18} atoms/cm^3)
(4) Lower band gap offers lower temperature range.	Higher band gap offers high temperature range.
(5) Wide applications	Limit applications.
(6) Electrons and holes value 3600	Electron and holes value 1700

Applications of semiconductor devices.

(i) Rectifier

(ii) Non. Linear resistors

(iii) Temperature sensitive resistors.

(iv) Photo-conductive and Photo Voltaic cells

- → Automatic control System
- → Television circuits
- → Sound, motion Picture recording.